

Samaranjan Manjhi

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SUMMARY

C/C++ Systems & Security Engineer with 2.5+ years of experience building production-grade endpoint security tools, kernel-level monitors, binary integrity frameworks, and DLP systems for macOS and Linux. Expertise in Endpoint Security framework, Ed25519 cryptographic signing, inotify/fanotify, system call hooking, and Qt. Delivered 6+ enterprise security products at MicroWorld Software Services.

TECHNICAL SKILLS

C, C++, QT, STL, DSA, SQL | Linux (Ubuntu/RHEL), macOS | Git, Xcode, Vim, GDB, Valgrind

PROFESSIONAL EXPERIENCE

Programmer (C/C++)

August 2023 – Present

MicroWorld Software Services Pvt Ltd

Andheri, Mumbai

- Engineered 6+ production security tools (binary integrity framework, DLP systems, FIM, content scanner, cloud backup) for macOS and Linux, deployed in enterprise endpoint security products.
- Developed kernel-level security solutions including a Linux `.ko` module for syscall interception and an Ed25519-based binary signing framework (`mw_guardian`) with inotify/fanotify enforcement.

KEY PROJECTS

`mw_guardian` | C/C++, inotify, fanotify, Ed25519, Linux Kernel (.ko)

Feb 2026 – Present

- Built a Linux binary integrity framework using inotify/fanotify to monitor watch directories; generates Ed25519 key pairs and embeds 64-byte signatures into binaries, shared libraries, and `.ko` modules at build time for tamper detection.
- Verifier binary validates signatures on execution — only watch-directory binaries/libraries require validation before CRUD operations; external binaries and syscalls bypass enforcement entirely, eliminating false positives.

Printer Control & Content Filter (DLP) | C/C++, Endpoint Security, libcup, Xcode

Oct 2025 – Jan 2026

- Built macOS DLP system intercepting print jobs at kernel level using the Endpoint Security framework; enforces application-based whitelisting/blacklisting and scans document content for PII (PAN, SSN, email) before allowing print operations.
- Implemented comprehensive audit logging of all blocked/allowed print events, enabling compliance reporting and forensic investigation of data exfiltration attempts via print.

File Integrity Monitoring (FIM) System | C/C++, OpenSSL, SQLite3

Jun 2025 – Aug 2025

- Configurable macOS FIM system storing MD5 hash, permissions, size, and timestamps in SQLite; achieves sub-second detection of unauthorized modifications with real-time alerting.
- Processes configuration-based watch lists and generates audit logs on every detected change, enabling forensic investigation and compliance reporting.

cBackup – Cross-Platform Cloud Backup & Monitor | C/C++, OpenSSL, cURL

Feb 2025 – May 2025

- Cross-platform (macOS/Linux) DLP agent that detects sensitive data being transferred over scp, rsync, USB, email, and network; triggers automatic local backup when a file containing PII or confidential data is shared.
- Creates AES-256 encrypted backup copies and synchronizes evidence to cloud at 10-minute intervals, ensuring data retention and audit trail for potential exfiltration incidents.

Content Scanner – DLP Engine | C/C++, Tesseract OCR, Poppler, Leptonica

Sep 2024 – Dec 2024

- PII detection engine (PAN, email, phone) across 12+ file formats (PDF, DOCX, images, ZIP, XML) via regex pattern matching and OCR for image-based documents.
- Packaged as both a shared library (`.so/.dylib`) for integration into security pipelines and a standalone CLI tool for on-demand compliance audits.

EDUCATION

Bachelor of Engineering in Computer Engineering

2019 – 2023

Shree LR Tiwari College Of Engineering, Mumbai University

CGPA: 8.56/10